

$$\begin{aligned}
(22a) \rightarrow & \frac{1}{16\pi^2} \left(\frac{1}{9} g_2^4 LF_{3,0}[m_2] + \frac{1}{6} g_2^4 LF_{4,-1}[m_2] - \frac{8}{45} g_2^4 LF_{5,-2}[m_2] - \right. \\
& \frac{1}{27} \sum_p g_1^4 LF_{3,0}[m_d^p] + \frac{5}{72} \sum_p g_1^4 LF_{4,-1}[m_d^p] - \frac{4}{135} \sum_p g_1^4 LF_{5,-2}[m_d^p] - \frac{1}{9} \sum_p g_1^4 LF_{3,0}[m_e^p] + \\
& \frac{5}{24} \sum_p g_1^4 LF_{4,-1}[m_e^p] - \frac{4}{45} \sum_p g_1^4 LF_{5,-2}[m_e^p] + \frac{1}{18} \sum_p (-g_1^4 + g_2^4) LF_{3,0}[m_l^p] + \\
& \frac{5}{48} \sum_p (g_1^4 - g_2^4) LF_{4,-1}[m_l^p] - \frac{2}{45} \sum_p (g_1^4 - g_2^4) LF_{5,-2}[m_l^p] - \frac{1}{54} \sum_p (g_1^4 - 9g_2^4) LF_{3,0}[m_q^p] + \\
& \frac{5}{144} \sum_p (g_1^4 - 9g_2^4) LF_{4,-1}[m_q^p] - \frac{2}{135} \sum_p (g_1^4 - 9g_2^4) LF_{5,-2}[m_q^p] - \\
& \frac{1}{27} \sum_p g_1^4 LF_{3,0}[m_u^p] + \frac{5}{18} \sum_p g_1^4 LF_{4,-1}[m_u^p] - \frac{16}{135} \sum_p g_1^4 LF_{5,-2}[m_u^p] + \\
& \frac{1}{18} (-g_1^4 + g_2^4) LF_{3,0}[\tilde{\mu}] + \frac{1}{12} (-g_1^4 + g_2^4) LF_{4,-1}[\tilde{\mu}] + \frac{4}{45} (g_1^4 - g_2^4) LF_{5,-2}[\tilde{\mu}] + \\
& \frac{1}{16} g_1^4 LF_{2,2,-1}[m_1, \tilde{\mu}] + \frac{1}{8} g_1^4 m_1^2 LF_{2,2,0}[m_1, \tilde{\mu}] - \frac{2}{3} g_2^4 LF_{2,1,0}[m_2, \tilde{\mu}] - \\
& \frac{17}{48} g_2^4 LF_{2,2,-1}[m_2, \tilde{\mu}] + \frac{1}{8} g_2^4 m_2^2 LF_{2,2,0}[m_2, \tilde{\mu}] + \frac{1}{3} g_2^4 LF_{3,1,-1}[m_2, \tilde{\mu}] + \\
& \frac{1}{3} g_2^4 LF_{3,2,-2}[m_2, \tilde{\mu}] - \frac{1}{6} g_1^2 (\overline{a_d^{pr}} a_d^{pr} + \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr}) LF_{2,2,0}[m_d^r, m_q^p] + \\
& \frac{1}{12} g_1^2 (\overline{a_d^{pr}} a_d^{pr} + \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr}) LF_{3,2,-1}[m_d^r, m_q^p] - \frac{1}{6} g_1^2 (\overline{a_e^{pr}} a_e^{pr} + \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr}) \\
& LF_{2,2,0}[m_e^r, m_l^p] + \frac{1}{12} g_1^2 (\overline{a_e^{pr}} a_e^{pr} + \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr}) LF_{3,2,-1}[m_e^r, m_l^p] + \\
& \frac{1}{6} g_1^2 (\overline{a_e^{pr}} a_e^{pr} + \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr}) LF_{3,1,0}[m_l^p, m_e^r] + \frac{1}{6} g_1^2 (\overline{a_e^{pr}} a_e^{pr} + \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr}) \\
& LF_{3,2,-1}[m_l^p, m_e^r] - \frac{1}{8} (3g_1^2 - g_2^2) (\overline{a_e^{pr}} a_e^{pr} + \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr}) LF_{4,1,-1}[m_l^p, m_e^r] + \\
& \frac{1}{6} (g_1^2 - g_2^2) (\overline{a_e^{pr}} a_e^{pr} + \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr}) LF_{5,1,-2}[m_l^p, m_e^r] + \frac{1}{6} g_1^2 (\overline{a_d^{pr}} a_d^{pr} + \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr}) \\
& LF_{3,1,0}[m_q^p, m_d^r] + \frac{1}{6} g_1^2 (\overline{a_d^{pr}} a_d^{pr} + \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr}) LF_{3,2,-1}[m_q^p, m_d^r] - \\
& \frac{1}{8} (5g_1^2 - 3g_2^2) (\overline{a_d^{pr}} a_d^{pr} + \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr}) LF_{4,1,-1}[m_q^p, m_d^r] + \\
& \frac{1}{2} (g_1^2 - g_2^2) (\overline{a_d^{pr}} a_d^{pr} + \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr}) LF_{5,1,-2}[m_q^p, m_d^r] + \\
& \frac{1}{12} (g_1^2 + 3g_2^2) (\overline{a_u^{pr}} a_u^{pr} + \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr}) LF_{2,2,0}[m_q^p, m_u^r] - \\
& \frac{1}{24} (g_1^2 + 3g_2^2) (\overline{a_u^{pr}} a_u^{pr} + \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr}) LF_{3,2,-1}[m_q^p, m_u^r] - \\
& \frac{1}{12} (g_1^2 + 3g_2^2) (\overline{a_u^{pr}} a_u^{pr} + \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr}) LF_{3,1,0}[m_u^r, m_q^p] - \\
& \frac{1}{12} (g_1^2 + 3g_2^2) (\overline{a_u^{pr}} a_u^{pr} + \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr}) LF_{3,2,-1}[m_u^r, m_q^p] - \\
& \frac{1}{4} (g_1^2 - 3g_2^2) (\overline{a_u^{pr}} a_u^{pr} + \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr}) LF_{4,1,-1}[m_u^r, m_q^p] + \\
& \frac{1}{2} (g_1^2 - g_2^2) (\overline{a_u^{pr}} a_u^{pr} + \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr}) LF_{5,1,-2}[m_u^r, m_q^p] + \\
& \frac{3}{2} \overline{y_d^{pr}} y_d^{sr} \overline{y_u^{st}} y_u^{pt} LF_{2,1,0}[m_u^t, m_d^r] - \frac{3}{2} \overline{y_d^{pr}} y_d^{sr} \overline{y_u^{st}} y_u^{pt} LF_{3,1,-1}[m_u^t, m_d^r] + \\
& \frac{1}{8} g_1^4 LF_{3,2,-2}[\tilde{\mu}, m_1] - \frac{1}{8} g_1^4 m_1^2 LF_{3,2,-1}[\tilde{\mu}, m_1] + \frac{5}{12} (g_1^4 - g_1^2 g_2^2) LF_{4,1,-2}[\tilde{\mu}, m_1] - \\
& \frac{1}{8} g_1^4 LF_{4,2,-3}[\tilde{\mu}, m_1] + \frac{1}{8} g_1^4 m_1^2 LF_{4,2,-2}[\tilde{\mu}, m_1] + \frac{1}{3} (-g_1^4 + g_1^2 g_2^2) LF_{5,1,-3}[\tilde{\mu}, m_1] + \\
& \frac{4}{3} g_2^4 LF_{3,1,-1}[\tilde{\mu}, m_2] + \frac{31}{24} g_2^4 LF_{3,2,-2}[\tilde{\mu}, m_2] - \frac{1}{8} g_2^4 m_2^2 LF_{3,2,-1}[\tilde{\mu}, m_2] + \\
& \frac{1}{4} g_2^2 (5g_1^2 - 9g_2^2) LF_{4,1,-2}[\tilde{\mu}, m_2] - \frac{5}{8} g_2^4 LF_{4,2,-3}[\tilde{\mu}, m_2] + \frac{1}{8} g_2^4 (m_2^2 + 4\tilde{\mu}^2) LF_{4,2,-2}[\tilde{\mu}, m_2] + \\
& (-g_1^2 g_2^2 + g_2^4) LF_{5,1,-3}[\tilde{\mu}, m_2] + 3g_1^2 g_2^2 \tilde{\mu}^2 LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] - \\
& 2g_1^2 g_2^2 \tilde{\mu}^2 LF_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] - \frac{3}{4} \overline{y_d^{st}} y_u^{sr} a_d^{pt} \overline{a_u^{pr}} LF_{2,2,1,-1}[m_q^p, m_u^r, m_d^t] + \\
& \frac{3}{4} y_d^{sr} \overline{y_u^{st}} (-\overline{a_d^{pr}} a_u^{pt} + \tilde{\mu}^2 \overline{y_d^{pr}} y_u^{pt}) LF_{2,2,1,-1}[m_q^p, m_u^t, m_d^r] + \\
& \frac{3}{4} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{sr} \overline{y_u^{st}} y_u^{pt} LF_{2,2,1,-1}[m_q^s, m_u^t, m_d^r] + \\
& \frac{3}{2} \overline{y_d^{st}} y_u^{sr} a_d^{pt} \overline{a_u^{pr}} LF_{2,1,1,0}[m_u^r, m_d^t, m_q^p] - \frac{3}{2} \overline{y_d^{st}} y_u^{sr} a_d^{pt} \overline{a_u^{pr}} LF_{3,1,1,-1}[m_u^r, m_d^t, m_q^p] + \\
& \frac{3}{2} y_d^{sr} \overline{y_u^{st}} (\overline{a_d^{pr}} a_u^{pt} - \tilde{\mu}^2 \overline{y_d^{pr}} y_u^{pt}) LF_{2,1,1,0}[m_u^t, m_d^r, m_q^p] + \\
& \frac{3}{2} y_d^{sr} \overline{y_u^{st}} (-\overline{a_d^{pr}} a_u^{pt} + \tilde{\mu}^2 \overline{y_d^{pr}} y_u^{pt}) LF_{3,1,1,-1}[m_u^t, m_d^r, m_q^p] - \\
& \frac{3}{2} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{sr} \overline{y_u^{st}} y_u^{pt} LF_{2,1,1,0}[m_u^t, m_d^r, m_q^s] + \\
& \frac{3}{2} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{sr} \overline{y_u^{st}} y_u^{pt} LF_{3,1,1,-1}[m_u^t, m_d^r, m_q^s] - \frac{1}{8} g_1^2 g_2^2 LF_{2,1,1,-1}[\tilde{\mu}, m_1, m_2] - \\
& \frac{1}{4} m_1 m_2 g_1^2 g_2^2 LF_{2,1,1,0}[\tilde{\mu}, m_1, m_2] - \frac{1}{4} g_1^2 g_2^2 LF_{3,1,1,-2}[\tilde{\mu}, m_1, m_2] + \\
& \frac{1}{4} g_1^2 g_2^2 (m_1 m_2 + 4\tilde{\mu}^2) LF_{3,1,1,-1}[\tilde{\mu}, m_1, m_2] + \frac{1}{4} g_1^2 g_2^2 LF_{4,1,1,-3}[\tilde{\mu}, m_1, m_2] - \\
& \frac{1}{4} g_1^2 g_2^2 (m_1 m_2 + 2\tilde{\mu}^2) LF_{4,1,1,-2}[\tilde{\mu}, m_1, m_2] - 2g_1^2 g_2^2 \tilde{\mu}^2 LF_{3,2,1,-2}[\tilde{\mu}, m_2, m_1] - \\
& \frac{1}{4} \tilde{\mu}^2 \over$$